

FIM 590 - HOMEWORK 1

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Question 1. Write an HJB equation for

$$\inf_{\pi} \mathbb{E} \left[\frac{\gamma}{2} \int_0^T \pi_s^2 + X_T^2 \right]$$

where

$$dX_t = -\pi_t dt + \sigma dW_t$$

with $X_0 = x \neq 0$ and $\gamma > 0$

Solve the HJB using the ansatz

$$H(t, x) = a(t)x^2 + b(t)$$

Solution. We have

$$H(t, x) = \sup_{\pi} \mathbb{E}^x \left[-\frac{\gamma}{2} \int_t^T \pi_s^2 - X_T^2 \right]$$

We have that the infinitesimal generator of X is

$$\mathcal{L}^{\pi} = \frac{1}{2} \sigma^2 \partial_x^2 - \pi_t \partial_x$$

Then we have:

$$\begin{cases} \frac{\partial H}{\partial t} + \sup_{\pi} (\mathcal{L}^{\pi} H - \frac{\gamma}{2} \pi_t^2) = 0 \\ H(T, x) = -x^2 \end{cases}$$

$$\frac{\partial H}{\partial t} + \sup_{\pi} \left(\frac{1}{2} \sigma^2 \partial_x^2 H - \pi_t \partial_x H - \frac{\gamma}{2} \pi_t^2 \right) = 0$$

The supremum is attained at

$$\pi_t = -\frac{1}{\gamma} \partial_x H$$

so we have

$$\begin{aligned} \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \partial_x^2 H + \frac{1}{\gamma} (\partial_x H)^2 - \frac{1}{2\gamma} (\partial_x H)^2 &= 0 \\ \frac{\partial H}{\partial t} + \frac{1}{2} \sigma^2 \partial_x^2 H + \frac{1}{2\gamma} (\partial_x H)^2 &= 0 \end{aligned}$$

Plugging in the ansatz, we have:

$$a'(t)x^2 + b'(t) + \sigma^2 a(t) + \frac{2}{\gamma} a(t)^2 x^2 = 0$$

which gives us the following ODEs:

$$\begin{cases} a'(t) + \frac{2}{\gamma} a(t)^2 = 0 \\ b'(t) + \sigma^2 a(t) = 0 \\ a(T) = -1, b(T) = 0 \end{cases}$$

We can solve the first ODE to get:

$$a(t) = -\frac{\gamma}{2(T-t) + \gamma}$$

and then solve the second ODE to get:

$$b(t) = -\frac{\sigma^2 \gamma}{2} \log \left(\frac{2(T-t)}{\gamma} + 1 \right)$$

and we have that the optimal control is:

$$\pi_t = \frac{1}{T-t + \gamma/2} X_t$$

to summarize:

$$\begin{cases} H(t, x) = -\frac{\gamma}{2(T-t) + \gamma} x^2 - \frac{\sigma^2 \gamma}{2} \log \left(\frac{2(T-t)}{\gamma} + 1 \right) \\ \pi_t = \frac{1}{T-t + \gamma/2} X_t \end{cases}$$

□

Question 2. Consider a risky asset price:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t$$

for a portfolio with self-financing condition:

$$dX_t^\pi = X_t^\pi \pi_t \frac{dS_t}{S_t} + r(1 - \pi_t) X_t^\pi dt - c_t dt$$

$c_t \geq 0$.

Find the optimal consumption and the optimal value for

$$\sup_{\pi} \mathbb{E} \int_0^\infty e^{-\rho t} \log(c_t) dt$$

where $\rho > 0$

Solution. The full dynamics for X_t^π are:

$$\begin{aligned} dX_t^\pi &= X_t^\pi \pi_t \mu dt + \sigma \pi_t X_t^\pi dW_t + r(1 - \pi_t) X_t^\pi dt - c_t dt \\ &= (X_t^\pi (\pi_t \mu + (1 - \pi_t) r) - c_t) dt + \sigma \pi_t X_t^\pi dW_t \end{aligned}$$

We have the following hamiltonian:

$$\mathcal{H}(t, x, p, M) = \sup_{\pi, c} \left\{ (x(\pi \mu + (1 - \pi) r) - c) p + \frac{1}{2} \sigma^2 \pi^2 x^2 M + e^{-\rho t} \log(c) \right\}$$

We can solve this as follows:

$$\begin{aligned} [\pi] : x(\mu - r)p + \sigma^2 \pi x^2 M &= 0 \\ [c] : -p + e^{-\rho t} c^{-1} &= 0 \end{aligned}$$

Which gives us the optimal controls:

$$\begin{cases} \pi = -\frac{p(\mu - r)}{\sigma^2 x M} \\ c = \frac{e^{-\rho t}}{p} \end{cases}$$

and gives us the following Hamiltonian:

$$\mathcal{H}(t, x, p, M) = xrp - \frac{p^2(\mu - r)^2}{2\sigma^2 M} - e^{-\rho t}(1 + \rho t + \log(p))$$

As such, we have the following HJB equation:

$$\begin{aligned} 0 &= \partial_t H + \mathcal{H}(t, x, \partial_x H, \partial_x^2 H) \\ &= \partial_t H + xr \partial_x H - \frac{(\partial_x H)^2}{\partial_x^2 H} \frac{(\mu - r)^2}{2\sigma^2} - e^{-\rho t} (1 + \rho t + \log(\partial_x H)) \end{aligned}$$

or the following PDE:

$$\frac{(\partial_x H)^2}{\partial_x^2 H} \frac{(\mu - r)^2}{2\sigma^2} + e^{-\rho t} (1 + \rho t + \log(\partial_x H)) = \partial_t H + xr \partial_x H$$

We think of this as follows. First, we guess that the value function is of the form $H(t, x) = e^{-\rho t} v(t, x)$, i.e. we guess that the value function is a discounted value function. This gives us the following PDE in v :

$$\frac{(\partial_x v)^2}{\partial_x^2 v} \frac{(\mu - r)^2}{2\sigma^2} + 1 + \rho v + \log(\partial_x v) = \partial_t v + xr \partial_x v$$

since we know that $H(t, x) \rightarrow 0$, and conceptualizing v as the undiscounted value, we guess that v reaches a steady state, so we guess that $\partial_t v = 0$, giving us the following ODE:

$$\frac{(v')^2}{v''} \frac{(\mu - r)^2}{2\sigma^2} + 1 + \rho v + \log(v') = xr v'$$

Additionally, we notice that the wealth should be scale invariant, i.e. the wealth and consumption should be independent of numeraire. This motivates the guess:

$$\begin{aligned} v(\lambda x) &= \sup_{\pi} \mathbb{E} \int_0^\infty e^{-\rho t} \log(\lambda c_t) dt \\ &= \frac{1}{\rho} \log(\lambda) + v(x) \end{aligned}$$

which, differentiating w.r.t. λ and setting $\lambda = 1$, gives us:

$$\begin{aligned} xv'(x) &= \frac{1}{\rho} \\ v(x) &= \frac{1}{\rho} \log(x) + C \end{aligned}$$

Plugging this into the ODE above gives us:

$$\begin{aligned} -\frac{(1/(\rho x)^2) (\mu - r)^2}{1/(\rho x^2)} \frac{(\mu - r)^2}{2\sigma^2} + 1 + \rho C - \log(\rho) &= \frac{r}{\rho} \\ -\frac{1}{\rho} \frac{(\mu - r)^2}{2\sigma^2} + 1 + \rho C &= \log(\rho) + \frac{r}{\rho} \\ C &= \frac{1}{\rho} \left(\log(\rho) + \frac{r}{\rho} + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right) \end{aligned}$$

We now verify that the candidate H satisfies the HJB equation. We begin with the left hand side:

$$\begin{aligned}
& \frac{(\partial_x H)^2}{\partial_x^2 H} \frac{(\mu - r)^2}{2\sigma^2} + e^{-\rho t}(1 + \rho t + \log(\partial_x H)) \\
&= \frac{\left(\frac{e^{-\rho t}}{\rho x}\right)^2}{-\frac{e^{-\rho t}}{\rho x^2}} \frac{(\mu - r)^2}{2\sigma^2} + e^{-\rho t} \left(1 + \rho t + \log\left(\frac{e^{-\rho t}}{\rho x}\right)\right) \\
&= \left(-\frac{e^{-\rho t}}{\rho}\right) \frac{(\mu - r)^2}{2\sigma^2} + e^{-\rho t}(1 + \rho t - \rho t - \log(\rho) - \log(x)) \\
&= e^{-\rho t} \left(-\frac{(\mu - r)^2}{2\rho\sigma^2} + 1 - \log(\rho) - \log(x)\right) \\
&= -e^{-\rho t} \left[\log(x) + \log(\rho) + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1\right]
\end{aligned}$$

Now for the right hand side

$$\begin{aligned}
\partial_t H + xr\partial_x H &= \frac{\partial}{\partial t} \left(\frac{e^{-\rho t}}{\rho} \left[\log(x) + \log(\rho) + \frac{(\mu - r)^2}{2\rho\sigma^2} + \frac{r}{\rho} - 1 \right] \right) + xr \left(\frac{e^{-\rho t}}{\rho x} \right) \\
&= -e^{-\rho t} \left[\log(x) + \log(\rho) + \frac{(\mu - r)^2}{2\rho\sigma^2} + \frac{r}{\rho} - 1 \right] + e^{-\rho t} \frac{r}{\rho} \\
&= -e^{-\rho t} \left[\log(x) + \log(\rho) + \frac{(\mu - r)^2}{2\rho\sigma^2} + \frac{r}{\rho} - \frac{r}{\rho} - 1 \right] \\
&= -e^{-\rho t} \left[\log(x) + \log(\rho) + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right]
\end{aligned}$$

Additionally, we see that $H(t, x) \rightarrow 0$ as $t \rightarrow \infty$, so indeed our H is the solution to the HJB equation. To summarize, we have that our value function is:

$$H(t, x) = \frac{e^{-\rho t}}{\rho} \log(x) + \frac{e^{-\rho t}}{\rho} \left(\log(\rho) + \frac{r}{\rho} + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right)$$

Now, we derive the optimal controls. We have that

$$\begin{aligned}
\pi_t^* &= -\frac{\partial_x H(\mu - r)}{\sigma^2 x \partial_x^2 H} \\
&= -\frac{\left(\frac{e^{-\rho t}}{\rho x}\right)(\mu - r)}{\sigma^2 x \left(-\frac{e^{-\rho t}}{\rho x^2}\right)} \\
&= \frac{\mu - r}{\sigma^2} \\
c_t^* &= \frac{e^{-\rho t}}{\partial_x H} = \frac{e^{-\rho t}}{\frac{e^{-\rho t}}{\rho x}} \\
&= \rho x
\end{aligned}$$

To summarize, we have the following value function and optimal controls:

$$\begin{cases} H(t, x) = \frac{e^{-\rho t}}{\rho} \log(x) + \frac{e^{-\rho t}}{\rho} \left(\log(\rho) + \frac{r}{\rho} + \frac{(\mu - r)^2}{2\rho\sigma^2} - 1 \right) \\ \pi_t^* = \frac{\mu - r}{\sigma^2} \\ c_t^* = \rho X_t \pi_t^* \end{cases}$$

□

Question 3. *Prove that Brownian motion has infinite total variation almost surely.*

Solution. We prove a stronger statement: Let M be a continuous local martingale. If M is almost surely non-constant, then M has infinite total variation almost surely.

Let $V_M(t, \omega)$ be the total variation of M on $[0, t]$. We assume for contradiction that $V_M(t) < \infty$ with positive probability. By stopping on the first time that $|M| > n$, we can assume that M is bounded. We then define

$$T_0^n := \inf\{t > 0 : |M_t| > 2^{-n}\}, T_k^n := \inf\{t > T_{k-1}^n : |M_t - M(T_{k-1}^n)| > 2^{-n}\},$$

and $t_k^n := T_k^n \wedge t$. Then, we have that

$$[M]_t^n := \sum_{k \geq 1} (M(t_k^n) - M(t_{k-1}^n))^2 \rightarrow [M]_t$$

uniformly a.s. This is true by Rogers and Williams IV.30.1. (I don't want to rewrite the proof here) However, we also have that

$$[M]_t^n \leq 2^{-n} \sum_{k \geq 1} |M(t_k^n) - M(t_{k-1}^n)| \leq 2^{-n} V_M(t)$$

So, since with positive probability, we have that $V_M(t) < \infty$, then with positive probability, we have $[M]_t = 0$ for $t > 0$. But, since

$$M^2 - [M]$$

is a martingale, stopping this martingale at

$$S := \inf\{t > 0 : [M]_t > 0\}$$

and optionally stopping, we have

$$\mathbb{E}[M_{S \wedge t}^2] = \mathbb{E}[[M]_{S \wedge t}] = 0$$

so that $M_{S \wedge t} = 0$, and since $S > 0$ with positive probability, We have that M is constant with positive probability, which is a contradiction.

Since Brownian Motion is a continuous local martingale, and is almost surely non-constant, we have that Brownian motion has infinite total variation almost surely. □